

Phase 1 Implementation Report

Vyapar AI — Universal AI Employee Platform

Status: **IMPLEMENTED**

Project: Z:\Vyapar AI\cursor

Date: June 2, 2026

Goal: Single-tenant MVP → multi-tenant-ready AI Employee foundation

Note: HONS is test tenant only — code is industry-agnostic

1. Executive Summary (संक्षिप्त)

Phase 1 successfully converts the codebase from hardcoded HONS chatbot logic into a **tenant-aware Universal AI Employee foundation**. All business data now flows through `company_manager.py` from `company_profiles.json`. Memory, Telegram, and OpenAI functionality are preserved.

2. Files Modified (7 + 1 doc)

File	Change
<code>company_profiles.json</code>	Generic tenant schema: industry, policies, rules, enriched products
<code>company_manager.py</code>	Central business abstraction layer (full rewrite)
<code>business_settings.py</code>	Loads from <code>company_manager</code> ; hardcoded HONS removed
<code>products.py</code>	Dynamic tenant catalog; generic <code>catalog_label</code>
<code>ai_employee_engine.py</code>	Tenant-aware via <code>COMPANY_ID</code> env
<code>main.py</code>	Startup validation + <code>company_loaded</code> on GET /
<code>env.example</code>	<code>COMPANY_ID</code> , <code>COMPANY_PROFILES_FILE</code> ; stale <code>BUSINESS_*</code> removed
<code>README_DEPLOY.md</code>	Tenant deployment notes

Unchanged (Preserved)

`memory.py`, `memory_db.py`, `memory_extractor.py`, `openai_engine.py`, Telegram webhook/polling, `intent_engine.py`, `knowledge_base.py`, `prompts.py`

3. Migration Notes

Topic	Detail
New env var	<code>COMPANY_ID=hons</code> (default if unset — backward compatible)
Optional	<code>COMPANY_PROFILES_FILE</code> overrides JSON path
JSON schema	New fields: <code>company_id</code> , <code>industry</code> , <code>currency</code> , <code>catalog_label</code> , <code>policies</code> , <code>rules</code> , <code>product tags/description</code>

Legacy JSON	Top-level installation_charge, router, onu still read via get_company_policies()
API change	business_context_to_prompt(company_id) requires company_id (internal only)
Backward compat	load_company() alias kept for get_company()
Multi-tenant	One COMPANY_ID per deployment instance (Phase 2 adds DB + admin)

4. company_manager.py — New API

```

CompanyProfileError
get_active_company_id()
get_company(company_id)
load_company(company_id)          # backward-compatible alias
require_company(company_id)
get_company_industry(company_id)
get_company_contact(company_id)
get_company_policies(company_id)
get_company_rules(company_id)
get_company_products(company_id)
get_catalog_label(company_id)
get_company_summary(company_id)
_load_all_profiles()              # isolated for future DB swap

```

5. File-by-File Diff Summary

company_profiles.json

- HONS reframed as test tenant (industry: "isp")
- Added policies, rules, product tags/description
- No HONS-specific logic in code — data only

company_manager.py

- Central business abstraction layer
- Product normalization: price formatting, auto-tags, duration
- Rules from JSON or auto-derived from policies/contact
- _load_all_profiles() isolated for Phase 2 DB migration

business_settings.py

- All 12 hardcoded constants removed
- business_context_to_prompt(company_id) builds generic business block

products.py

- Static PRODUCTS list removed
- Tenant-scoped search + prompt with generic catalog_label

ai_employee_engine.py

- generate_employee_reply(..., company_id=None) tenant-aware

- Memory/direct recall works even if tenant misconfigured
- Config error returned before OpenAI call on missing profile

main.py

- Startup: COMPANY_PROFILE_LOADED or COMPANY_PROFILE_MISSING log
- GET / returns company_id and company_loaded

6. Architecture After Phase 1

```
COMPANY_ID (env)
  ↓
company_profiles.json ← SSOT (Phase 1)
  ↓
company_manager.py ← Central Business Abstraction Layer
  ↓           ↓
business_settings products
  ↓           ↓
ai_employee_engine.py
├─ memory.py (unchanged)
├─ intent_engine + knowledge_base
└─ openai_engine.py (unchanged)
  ↓
Telegram (main.py)
```

Future-compatible hooks:

- `_load_all_profiles()` → DB repository (Phase 2)
- `company_id` param → multi-channel routing (Phase 3+)
- `get_company_rules()` → workflow engine input
- `get_company_products()` → CRM / lead engine catalog

7. Error Handling

Scenario	Behavior
COMPANY_ID missing	Defaults to "hons"
company_profiles.json missing	CompanyProfileError → user config error message
Unknown COMPANY_ID	Same — memory/direct recall still works
Company with no products	Prompt: "None confirmed in company profile"
Company with no rules	Auto-derived from policies, support hours, contact

8. Testing Checklist

#	Test	Expected
1	COMPANY_ID=hons + pricing question	Products from JSON in reply
2	COMPANY_ID=invalid	"not configured yet" message
3	GET /	company_id, company_loaded: true/false
4	Startup logs	COMPANY_PROFILE_LOADED or MISSING
5	"Mero naam X ho" → "Mero naam ke ho?"	Memory works without tenant block
6	Telegram webhook/polling	Unchanged behavior
7	GET /healthz	Database healthy
8	Add 2nd tenant + new COMPANY_ID deploy	Different business context

9. Rollback Plan

1. **Git:** Revert modified files to previous commit
2. **Render:** Remove COMPANY_ID env var (old code ignored it)
3. **No DB migration** — memory schema unchanged; rollback is code-only
4. Keep JSON backup before adding new tenants

10. Technical Debt Remaining

Item	Target Phase
JSON file as SSOT (no hot reload)	Phase 2 → DB
No company_id on memory tables	Phase 2
Sync SQLAlchemy in async path	Phase 2
Blocking webhook handler	Phase 2
company_profile.json duplicate file	Delete when ready
No automated tests	Phase 2
Keyword-only product/KB search	Phase 3+
No lead/workflow/CRM layers	Phase 3

11. Recommended Phase 2

1. **DB tenant model** — companies, company_products, company_knowledge tables
2. **Repository pattern** — swap `_load_all_profiles()` without changing public API
3. **company_id on memory** — tenant-scoped user data
4. **Alembic migrations**
5. **Admin API** (authenticated CRUD — not full panel)
6. **Background worker** — async webhook processing
7. **Second test tenant** — e.g. hospital_demo industry seed

12. Quick Start

```
COMPANY_ID=hons
```

Add new tenant: new key in `company_profiles.json` → deploy with matching `COMPANY_ID`.

Strategic note: HONS is test data only. All code paths are tenant-generic and industry-agnostic. Platform supports ISP, Hospital, School, Travel, Real Estate, Retail, Banking, Insurance via JSON config — not code changes.